

# **A Critical Study of Misconfiguration-Driven Security and Forensic Risks in Web-Based POS Deployments for SMEs**

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## Abstract

This study critically examined misconfiguration-driven security and forensic risks in web-based point-of-sale (POS) deployments used by small and medium enterprises (SMEs). It focused on how realistic configuration weaknesses may lead not only to exploitable vulnerabilities, but also the persistence and recoverability of sensitive transactional data across client-side, server-side, and network artefact sources.

A review of existing literature was conducted, covering SME cyber security constraints, formal frameworks and standards – including NCSC, OWASP, and PCI SSC guidance – and digital forensic artefact persistence. These findings informed the development of an analytical framework linking misconfiguration types to likely artefact locations and security consequences.

The analytical framework was then empirically validated using a controlled open-source POS test environment. Five selected misconfiguration scenarios were implemented: debug mode and verbose logging, insecure transport, client-side session storage misuse, lack of encryption at rest, and improper data invalidation. Each scenario produced recoverable residual artefacts in at least one system layer, with synthetic payment data recovered from logs, network traffic, browser storage, database storage, and underlying storage artefacts.

The findings demonstrate that misconfiguration is a practical and high-impact risk for SME POS environments, particularly where encryption, secure deletion, and lifecycle controls are weak or inconsistently implemented. The study contributes a structured framework for understanding the relationship between misconfiguration, data remanence, and forensic recoverability.

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## List of Abbreviations

| <u>Abbreviation</u> | <u>Definition</u>                                |
|---------------------|--------------------------------------------------|
| ASVS                | Application Security Verification Standard       |
| CWE                 | Common Weakness Enumeration                      |
| IDS                 | Intrusion Detection System                       |
| LAMP                | Linux, Apache, MySQL, PHP                        |
| NCSC                | National Cyber Security Centre                   |
| OSPOS               | Open Source Point of Sale                        |
| OWASP               | Open Worldwide Application Security Project      |
| PAN                 | Primary Account Number                           |
| PCI DSS             | Payment Card Industry Data Security Standard     |
| PCI SSC             | Payment Card Industry Security Standards Council |
| POS                 | Point of Sale                                    |
| RQ                  | Research Question                                |
| SAD                 | Sensitive Authentication Data                    |
| SIEM                | Security Information and Management System       |
| SME                 | Small and Medium Enterprise                      |
| VM                  | Virtual Machine                                  |

# Chapter 1. Introduction

## 1.1. Background and Context

Small and Medium Enterprises (SMEs) are increasingly adopting low-cost, web-based point-of-sale (POS) systems to support transaction processing and day-to-day operations, driven by broader digital transformation trends and the cost-efficiency and flexibility of these solutions (Pagidoju, 2026). These web applications are often built on standard web architectures, but such deployments are frequently carried out by individuals with limited technical expertise, increasing the likelihood of configuration-related issues.

Given that POS systems process sensitive transactional data, even minor configuration errors can result in significant security, compliance, and financial consequences. In this context, the security of SME POS deployments is not solely dependent on system design, but also on how systems are configured and maintained in practice.

## 1.2. Problem Statement

Existing security frameworks and standards, including Cyber Essentials, OWASP guidance, and PCI SSC guidance, provide organisations with high-level principles and requirements for secure system design and operation. However, this guidance is often technologically-agnostic and does not always translate effectively into practical implementation within SME environments.

As a result, SMEs may struggle to apply these requirements within their specific technical contexts, increasing the likelihood of misconfiguration. Such misconfigurations can lead not only to actively exploitable vulnerabilities, but also to the unintended persistence of sensitive data across system components. This creates an additional layer of risk, where sensitive information may be recoverable through forensic analysis, even in the absence of a direct security breach.

## 1.3. Study Purpose

### 1.3.1. Aim

To critically examine how realistic misconfigurations in web-based point-of-sale deployments used by SMEs create security and forensic risk, and to evaluate the extent to which such risks are addressed by existing security frameworks, supported by empirical validation.

### 1.3.2. Questions

RQ1: What misconfiguration-related security risks are most prominent in web-based POS deployments used by SMEs?

RQ2: How do existing guidance and compliance frameworks, including Cyber Essentials, OWASP, and PCI DSS, address these risks and where do practical gaps remain for SMEs?

RQ3: To what extent do selected misconfiguration scenarios lead to recoverable residual artefacts in a controlled open-source POS test environment?

### 1.3.3. Objectives

Ob1: To identify and classify common misconfiguration-related security risks in web-based POS deployments used by SMEs, based on recent academic literature and industry guidance.

Ob2: To critically evaluate how existing frameworks, including Cyber Essentials, OWASP, and PCI DSS, address misconfiguration-related risks, and to identify practical limitations in their application within SME environments.

Ob3: To design and implement a controlled experiment testbed using an open-source POS system in order to assess the extent to which selected misconfigurations result in recoverable residual data artefacts.

Ob4: To analyse the relationship between misconfiguration, data persistence, and forensic recoverability, and to evaluate the resulting security and compliance implications for SMEs.

## 1.4. Scope and Delimitations

This study focuses on web-based POS systems operating within a client-server architecture, as these are commonly adopted by SMEs due to their low cost and scalability. The research specifically examines misconfiguration-driven risks and their impact on data persistence, rather than broader threat categories such as malware, insider threats, or open-source software policies.

The empirical validation is conducted using a single open-source POS platform selected to reflect realistic SME deployment scenarios, while maintaining transparency and control within the experimental environment. Proprietary systems are excluded due to accessibility and ethical constraints. While this approach enables detailed analysis, it limits the generalisability of findings across all POS implementations.

## 1.5. Paper Structure

This study is structured into seven chapters. Chapter 2 reviews existing literature, examining SME security challenges, common misconfiguration risks, data persistence mechanisms, and relevant security frameworks. Chapter 3 introduces the analytical framework, including the Misconfiguration-to-Artifact and Remanence Source models that underpin the study. Chapter 4 outlines the research methodology and experimental design. Chapter 5 presents the results of the empirical validation, Chapter 6 discusses these findings in relation to the research questions, and Chapter 7 concludes the study.



## Chapter 2. Review of Existing Literature

### 2.1. Introduction

This chapter reviews existing literature relevant to misconfiguration-related security risks in web-based point-of-sale (POS) systems used by SMEs. It examines the role of web-based POS architectures, the technical and organisational factors that contribute to misconfiguration, and the nature of data persistence and forensic artefacts within the POS environments. In addition, it evaluates existing security frameworks in order to assess how effectively they address these risks in practice. The chapter concludes by identifying key gaps in the literature, which inform the analytical framework and empirical validation chapters.

### 2.2. Web-Based POS Architecture and Characteristics

Web-based POS systems commonly use a client-server architecture, in which browser-based clients send transaction data to a server-side application and database. This model is often built on a collection of layered software called a 'software stack', such as the LAMP stack, as visualised in Figure 1. This model is attractive to SMEs as it can be deployed using low-cost infrastructure and low-resource thin client devices. However, this architecture distributes data across multiple layers: client browsers, server logs and databases, and network traffic. This creates several potential locations for residual artefacts when systems are misconfigured.

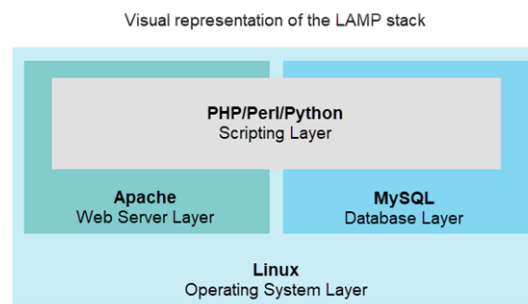


Figure 1 – LAMP Stack Architecture (Kaplarevic, 2024)

### 2.3. SME Constraints and Maturity Challenges

Security misconfigurations, while technical in nature, frequently derive from an often-overlooked issue involving the constraints and challenges faced by SMEs. Financial and resource constraints are often cited as the primary limitations, but SME cyber security weaknesses are also driven by lack of expertise, poor risk perception, and difficulty interpreting and implementing available guidance (Chidukwani, Zander and Koutsakis, 2022).

Despite their economic importance – making up 99.85% of all business in the UK (Department for Business & Trade, 2025) – existing research remains limited and narrowly focused for SMEs, indicating a gap in the academic understanding of SME cyber security practices. Furthermore, in spite of the availability of established protective measures, SMEs have consistently shown to under-implement existing cyber security controls, resulting in continued vulnerability to cyber-attacks (Chidukwani, Zander and Koutsakis, 2022).

The Department for Science, Innovation & Technology (2025) found that more small businesses were implementing cyber hygiene practices such as risk assessments and policies. However, as noted, these practices are still difficult for many SMEs to interpret and implement into their environments, hence why cyber governance and monitoring continue to be weaker than medium and large organisations in comparison, decreasing the likelihood of systematic secure configurations.

SMEs generally lack formal security governance, dedicated security staff, and standardised configuration baselines (Awan, Alam, and Kamran, 2025). As such, they are often reactive rather than proactive, prioritising usability, with security issues often addressed in response to incidents. Furthermore, inconsistent system configurations, weak policies, and lack of auditing can cause outdated configurations to remain undetected and sensitive data left exposed.

Overall, cyber security misconfigurations can be derived from the constraints and challenges faced by SMEs, including lack of implementation-focused guidance, policies, audits, and technical expertise, significantly increasing the risk of misconfigurations and other poor security practices.

## 2.4. Formal Guidance and Compliance Frameworks

Given these constraints, SMEs often turn to formal guidance and compliance frameworks for support in the absence of technical expertise. However, existing guidance is typically designed to provide broad, technology-agnostic recommendations rather than detailed technical controls tailored to specific web models. As a result, the applicability of such frameworks to open-source, web-based POS solutions remains unclear.

### 2.4.1. NCSC Cyber Essentials

While the availability of formal technical guidance appears to be limited, one of the most prolific starting points is a certification by the NCSC named Cyber Essentials. It succeeds in providing support for small business by covering the essentials of cyber security hygiene, with training and a certification. However, this framework is broadly focused on technical infrastructure rather than organisational-specific application hardening. The framework states that the requirements apply to all devices and software that control the flow of data across the internet (NCSC Cyber Essentials, 2025, p.7), which applies to web-based POS systems, as they typically require internet connectivity to exchange transactional data with banks.

However, given that Cyber Essentials covers a range of basic vulnerabilities with beginner-friendly information, where 'secure configurations' involve preliminary steps, such as changing default passwords and securing firewalls, this framework is designed to help prevent common internet-based cyber-attacks and does not address data persistence risks. Therefore, Cyber Essentials is not sufficient as a standalone form of configuration-focused guidance for SMEs.

### 2.4.2. OWASP

Across several layers of the software stack, web applications are susceptible to a multitude of security risks and vulnerabilities. The OWASP Top 10 (OWASP, 2025b) categorises these risks primarily in terms of actively exploitable weaknesses, which can be used alongside the Application Security Verification Standard (ASVS) (OWASP, 2025a), which provides a set of verification requirements to translate these risks into practical controls.

The OWASP Top 10 is a community-based, data-driven list of the most critical web-application security risks. The categories within the Top 10 map to several associated Common Weakness Enumerations (CWEs), which are a catalogue of software and hardware weaknesses that have the potential to develop into vulnerabilities (MITRE, no date).

Across the categories within the Top 10, there is a pattern in which vulnerabilities that are traditionally associated with active exploitation can also contribute to unintended data persistence when misconfigured. For example, 'A04 – Cryptographic Failures' can impact POS systems in which unencrypted customer details and sensitive payment information is processed in plaintext, this data may be recoverable. This shifts the risk from immediate compromise to the prolonged exposure of sensitive data.

Furthermore, other categories can present SME challenges, for example, 'A09 – Security Logging and Alerting Failures' includes insufficient logging (CWE-778) can hinder monitoring and response, whereas excessive logging (CWE-779) may store sensitive information (CWE-532). Achieving this balance of detailed yet minimal logs may present significant challenges to SMEs. Lastly, 'A02 – Security Misconfiguration' reflects the increasing importance of configuration-based weaknesses, and includes debug mode, (CWE-489), improper logging configuration (CWE-532), and misconfigured storage permissions (CWE-16).

While the Top 10 identifies common categories of risk, it does not provide detailed implementation guidance. To supplement this, OWASP ASVS provides a structured set of verification requirements and can be used to translate high-level risks into practical controls.

Collectively, OWASP provides a list of common vulnerabilities, alongside a comprehensive and formal framework that SMEs can use as a review checklist when deploying web applications. However, a non-technical SME user would struggle to take ASVS alone and secure a web application, therefore the likelihood of potential misconfigurations remains high.

### 2.4.3. PCI DSS

Moving from technical guidance to security compliance, it is necessary to consider the security standards that organisations processing card transactions are required to adhere to. The Payment Card Industry Security Standards Council (PCI SSC) provide a contractually mandatory standard for any entity handling cardholder data, named the PCI Data Security Standard (PCI DSS).

The Requirements and Testing Procedures (PCI DSS, 2024) defines the official PCI DSS standard, with guidance covering what merchants must implement and how assessors verify compliance. Furthermore, it defines what counts as cardholder data and outlines storage requirements.

PCI DSS states that 'account data' consists of the full Primary Account Number (PAN), cardholder data present alongside the PAN, and any elements of Sensitive Authentication Data (SAD). Figure 2 illustrates the elements of cardholder data and SAD alongside their storage requirements.

PCI DSS requires PAN storage kept to a minimum and protected behind strong cryptographic algorithms, which is not the case for other cardholder data elements. It also requires that this data must be *securely* deleted when no longer needed. Furthermore, SAD must *not* be stored after authorisation under any circumstances, even if the data is encrypted; it must be protected in-memory and securely deleted during temporary handling.

|              |                               | Data Elements                | Storage Restrictions                                                              | Required to Render Stored Data Unreadable                                                                                   |
|--------------|-------------------------------|------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Account Data | Cardholder Data               | Primary Account Number (PAN) | Storage is kept to a minimum as defined in Requirement 3.2                        | Yes, as defined in Requirement 3.5                                                                                          |
|              |                               | Cardholder Name              | Storage is kept to a minimum as defined in Requirement 3.2 <sup>2</sup>           | No                                                                                                                          |
|              |                               | Service Code                 |                                                                                   |                                                                                                                             |
|              |                               | Expiration Date              |                                                                                   |                                                                                                                             |
|              | Sensitive Authentication Data | Full Track Data              | Cannot be stored after authorization as defined in Requirement 3.3.1 <sup>3</sup> | Yes, data stored until authorization is complete must be protected with strong cryptography as defined in Requirement 3.3.2 |
|              |                               | Card verification code       |                                                                                   |                                                                                                                             |
|              |                               | PIN/PIN Block                |                                                                                   |                                                                                                                             |

Figure 2 – Account Data Storage Requirements (PCI DSS, 2024, p.5)

Each PCI DSS requirement contains substantial advice for good practice and resources for further guidance. For example, SAD storage guidance provides a list of potential data sources to review, such as debug logs. However, while this represents a highly valuable and robust source of guidance, it is important to recognise that the availability of such information does not directly determine an SME’s ability to be compliant.

Furthermore, these requirements can present difficulties to SMEs, for example, Requirement 2 requires secure configurations of systems, such as removing default accounts and applying secure configuration baselines, but SMEs may rely on vendor defaults, particularly when deploying self-hosted systems without sufficient expertise (Chidukwani, Zander and Koutsakis, 2022), and Requirement 4 requires transmitted cardholder data to be encrypted, but implementing secure encryption can be technically demanding for SMEs.

Kaur, R. (2020) studied the PCI DSS challenges faced by SMEs and draws on Rees, J. (2012) that there is no difference in PCI DSS compliance, independent of company size. Kaur also highlights that SME challenges extend financial and resource constraints, and lack of awareness of these compliance requirements entirely. Building on this observation, small firms may deploy these payment systems and, because they don’t physically see the transactional data, may not question its security, much less the existence of these requirements they are contractually obliged to meet.

PCI DSS provides comprehensive security requirements rather than technical guidance, proving to be a beneficial information source for SMEs. However, these requirements may not always be widely understood or even recognised by smaller organisations, and achieving compliance may continue to present technical and operational challenges to SMEs, illustrating the gap between guidance and practical implementation.

2.4.4. SME Guidance Evaluation

This section has identified four sources of formal guidance available to SMEs. Cyber Essentials provides a broad baseline and certification for general cyber security hygiene practices, OWASP offers a risk prioritisation reference point alongside security controls, and PCI DSS provides both compliance and control measures.

While no single source provides full guidance coverage, each offers different guidelines and approaches. Therefore, the best alternative for an SME would be to combine all four frameworks, as this would bridge their respective gaps. Together, these can provide broad cyber security understanding, certifications, compliance requirements, and control checklists. However, this approach is still not optimal, as the different forms of information may introduce complexity and overburden small businesses. In conclusion, the available guidance proves to be useful but remains insufficient.

## 2.5. Misconfiguration as a Security Risk

Misconfiguration represents a significant source of security risk within web applications and is often overlooked in research into SME technical environments. Unlike software vulnerabilities, misconfigurations arise from the way systems are configured and maintained. However, these are derived from various underlying factors, and this section examines the common causes of misconfiguration and analyses how each weakness may result in the unintended persistence of sensitive data.

### 2.5.1. Human Error and Insecure Defaults

Human error is one of the primary root causes of misconfigurations. In most cases, particularly in SME environments, this comes down to lack of expertise, misunderstanding documentation, and reliance on default configuration.

A common mistake includes debug mode being enabled in deployed systems, which is a mode designed to allow developers to monitor, test, and troubleshoot systems, often providing detailed access to logs, memory, and execution processes (Gallagher, 2025). Other mistakes may include excessive file permissions and poor database access controls, which can both expose transactional data being written to logs, temporary files, or debugging outputs, where it persists beyond the intended processing window.

Default configurations are another common cause of misconfigurations for SMEs, which are the initial configuration states of systems prior to customisation. Many of these settings are designed to be reconfigured by the user before deployment, but these settings may remain unchanged in SME environments for a multitude of reasons, such as the prioritisation of usability, misconceptions around default security, or even to avoid *introducing* misconfigurations, typically down to their lack of expertise (Chidukwani, Zander and Koutsakis, 2022). For example, Apache, the web server in the LAMP stack, can be deployed with HTTP by default, unless HTTPS is explicitly configured.

The Microsoft Power Apps exposure incident, in which default platform configurations caused the exposure of millions of sensitive organisational and governmental records (UpGuard, 2021), demonstrates how sensitive data can be exposed without any active exploitation, highlighting the risks associated with insecure default settings.

### 2.5.2. Lifecycle Management and Retention Failures

Poor lifecycle management represents another serious concern, as outdated configurations and redundant data are rarely revisited and sufficiently cleansed, potentially resulting in the prolonged retention of sensitive data in logs and backups.

Typical misconfiguration issues may include legacy configurations, in which obsolete configurations, even those that were previously secure, become outdated as system requirements and threat landscapes evolve. Additionally, mismanagement of backups and logs, both of which may never be cleaned or rotated, may contain historical records of transactions, further exacerbated by improper data invalidation, in which data can be marked as 'deleted', but remains forensically recoverable (Pratyashrit, Sharma and Sathiyasuntharam, 2026).

### 2.5.3. Convenience Over Security

SMEs often prioritise operational efficiency and business continuity over security considerations, particularly where resources and expertise are limited (Ejaz and Matthew, 2024; Chidukwani, Zander and Koutsakis, 2022). Controls that introduce friction, such as frequent re-authentication, may be weakened or disabled to simplify day-to-day use, and insecure configurations, such as storing application data within the client-side browser storage to reduce re-authentication prompts, may be implemented. Furthermore, in attempts to deploy these features, SMEs may unknowingly implement insecure code without understanding exactly how it works.

While these decisions are intended to improve usability, they can also increase the persistence of sensitive data, as long-lived sessions and data may remain accessible client-side long-term. The balance between usability and security presents a significant challenge for SMEs seeking simple and efficient workflows.

Collectively, these sources of misconfiguration must be recognised when considering the impact they have on SME environments, as misconfiguration is the underlying factor connecting security failure and forensic persistence.

## 2.6. Data Persistence and Forensic Artefacts

In web-based environments, data can persist across multiple layers, including client-side, server-side, and network layers. Artefacts can reside across volatile memory, filesystems, logs, backups, and network traffic (FFaizal and Luthfi, 2024). Understanding these sources of data persistence provides a structured basis for examining the remanence of transactional data after POS use.

### 2.6.1. Server-side Artefacts

The primary source of data remanence is from within the server, as it can handle several types of information during web hosting, application operation, and database management. Web and app logs, for example, may store events such as webapp logons, transaction details, and database updates, but poor logging configurations may cause the inclusion of sensitive payment data and credentials. Potential server-side data remanence source locations include logs, databases, backup systems, and temporary files.

### 2.6.2. Client-side Artefacts

Client-side systems may contain residual artefacts from web application usage due to session storage and caching mechanisms, making it a valuable source of residual data. Browsers may leave artefacts across local storage, logs, and system-level traces throughout their lifecycle (Raza *et al.*, 2024). Session storage, for example, typically stores data temporarily for use within the same session, such as form inputs, but mishandling of this storage could inadvertently result in the storage of payments inputs.

### 2.6.3. Network Artefacts

Network artefacts may expose sensitive transactional data where insecure transmission mechanisms are used. HTTPS, for example, ensures that captured network packets are cryptographically secured. However, misconfigured web servers may accept HTTP traffic, allowing sensitive data to be transmitted in plaintext. This potential data leakage is not only vulnerable to intercepted network traffic, but environments with Security Information and Event Management (SIEM) and Intrusion Detection Systems (IDS) may store this data beyond the transmission alone.

## 2.7. Literature Synthesis and Research Gaps

While the importance of small businesses is acknowledged, existing literature remains limited relative to the overall volume of SMEs. While the literature discusses SME security, POS security, web application risks, compliance standards, and forensic artefacts separately, there are few studies that comprehensively connect them together. This section has identified a gap in which there is no work linking realistic SME misconfigurations to specific residual artefacts in web-based point-of-sale systems, even with the existence of real-world cyber incidents caused by web application misconfigurations.

This study addresses this identified gap by selecting common, realistic misconfiguration scenarios, such as verbose logging, cryptographic failures, and insecure defaults, identifying where residual sensitive data artefacts reside, and discussing the consequences this could have to an SME environment and why further research is needed for this area.

## Chapter 3. Analytical Framework

### 3.1. Introduction

This chapter presents the analytical framework used to guide the empirical validation. The framework provides a structured way of linking common SME-relevant POS misconfigurations to their likely security consequences and residual artefact locations, establishing a basis for predicting where sensitive data may persist within a POS environment. This framework is expanded upon in [Appendix D – Expanded Analytical Framework Tables](#)

### 3.2. Misconfiguration Matrix Model

Table 1 summarises each identified misconfiguration, mapped to its potential manifestation within a POS environment and its potential impact on data persistence and forensic recoverability.

*Table 1 – Misconfiguration Matrix*

| Misconfiguration                          | Applied in POS Environment                                            | Potential Impact                                                                                   |
|-------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Debug mode enabled in production          | POS web application deployed with debugging active                    | Logs and error outputs may contain sensitive transactional and cardholder data.                    |
| Verbose logging                           | Logging configured to capture full request/response data              | Logs may contain sensitive transactional and cardholder data.                                      |
| Default configurations                    | POS deployed without modifying default settings                       | Implicit enabling of insecure behaviours such as logging, weak access controls, and data retention |
| Lack of encryption at rest                | Sensitive data stored long-term not cryptographically secured         | Plaintext data can be directly read from storage                                                   |
| Improper data invalidation                | Data marked as deleted but not overwritten to ensure complete removal | Underlying storage and unallocated space may contain residual data                                 |
| Backup mismanagement                      | Backups created without encryption or retention limits                | Long-term storage of sensitive data in backup files                                                |
| Inconsistent configuration across devices | Different POS terminals configured differently                        | Unpredictable data storage locations, complicating detection and increasing persistence risk       |
| Session Storage Misuse                    | Browser Local/Session storage contains transactional data             | Sensitive data may persist client-side beyond intended use                                         |
| Weak access controls                      | Poorly configured database permissions                                | Sensitive data accessible to unauthorised users and processes                                      |
| Insecure transport                        | POS system transmits transaction details over HTTP rather than HTTPS  | Plaintext data can be intercepted or stored through monitoring systems                             |

Collectively, these misconfigurations demonstrate how non-malicious configuration practices can create conditions in which sensitive payment data may persist across various system components, forming the foundation of this study's empirical validation.



### 3.3. Remanence Source Model

The potential locations of data remanence in a web-based POS system are summarised in Table 2. Each source is mapped to its method(s) of examination and expected data types.

Table 2 – Remanence Sources

| Artefact Source       | Location in POS Environment | Examination Method                       | Potential Recoverable Data                  |
|-----------------------|-----------------------------|------------------------------------------|---------------------------------------------|
| Browser Cache         | Client-side                 | Browser forensic tools, cache inspection | Cached transaction pages, form inputs       |
| Local Session Storage | Client-side                 | Developer tools, storage inspection      | Temporary transaction data                  |
| Cookies               | Client-side                 | Cookie analysis tools                    | Session IDs, user data                      |
| Application Logs      | Server-side                 | Log file analysis                        | Transaction records, PANs, debug data       |
| Web Server Logs       | Server-side                 | Log file analysis                        | Request/Response data                       |
| Database Storage      | Server-side                 | SQL queries, database dumps              | Stored transaction records, cardholder data |
| Temporary Files       | Server-side                 | File system analysis                     | Intermediate transaction data               |
| Backups               | Server-side                 | Backup file analysis                     | Historical transaction data                 |
| Network Packets       | Network Layer               | Packet sniffing tools                    | Possible plaintext payment data             |
| Memory                | Client-side or Server-side  | Memory forensic tools                    | Live transaction data, live session data    |

These artefact sources demonstrate that sensitive transactional data may persist across multiple layers of a web-based POS deployment. The distribution of data across client-side, server-side, and network environments provides a structured framework for forensic investigation and the empirical validation.

### 3.4. Chapter Summary

This analytical framework provides the foundation for the empirical validation conducted in this study. By linking misconfigurations to their potential applications within POS environments with expected artefact locations and likely consequences, this framework allows each scenario to be examined systematically. Furthermore, it provides a basis for discussing how common SME misconfigurations may contribute to both security and compliance risk.

## Chapter 4. Methodology

### 4.1. Secondary Research Design

Academic literature was identified through Google Scholar, IEEE Xplore, ACM Digital Library, ScienceDirect.

Search terms combined 'POS', 'SME', 'Open-source'/'Web-based Systems', 'PCI DSS', 'OWASP', 'Cyber Essentials', and 'Forensic Artefact Recovery'.

The criteria for papers included:

- Date of publication – within the last six years, to ensure real-world relevance while not immediately excluding papers of interest.
- Relevance – relative to the topic at hand, but generally relating to open-source, web-based POS solutions and digital forensic artefact recovery.
- Unbiased – ensuring that any non-study-based research does not have any underlying partiality, particularly those designed to advertise a product or service.

Findings from the secondary research were used to design the empirical validation, enabling selected risks identified in the literature to be examined in a controlled environment.

### 4.2. Selection of Standards and Guidance

NCSC Cyber Essentials (2025) and OWASP ASVS (2025a) were used to investigate current available technical and security guidance for SMEs. The former chosen as it is a UK Government Organisation that provides certifications, and the latter chosen as it is a widely recognised formal framework.

OWASP Top 10 (2025b) was used to identify common categories of web application risks, several of which may arise from misconfigurations in SME POS environments.

PCI DSS (2024) was used as it is a global security framework for entities handling cardholder data, in which compliance is contractually mandatory, providing a reference point for SME compliance challenges.

### 4.3. Testbed Justification

The standard LAMP stack environment was designed with an Ubuntu Server virtual machine to act as a lightweight web server and database, using Apache and MySQL respectively, as well as PHP for the scripting layer. A Windows 10 'thin client' virtual machine was designed to simulate a POS terminal, connected to the Ubuntu Server via LAN to access the web application.

#### 4.3.1. POS Platform

The web app used was Open-Source Point-of-Sale (OSPOS), which is an open-source, web-based POS system, representative of low-cost solutions available to SMEs. While it is not the only platform within this category, it provides an appropriate foundation for this study due to its accessibility, transparency, and flexibility. As an open-source application, OSPOS allows full visibility and modification of its underlying codebase, enabling controlled configuration and the intentional introduction of misconfigurations required for experimental analysis. This level of access would not typically be possible with proprietary POS systems.

Furthermore, its web-based architecture, built on a standard LAMP stack, reflects common deployment practices within SME environments, making it a suitable proxy for examining real-world security weaknesses and data persistence behaviours. The use of OSPOS therefore supports a controlled, ethical, and reproducible experimental environment for analysing the lifecycle and remanence of sensitive data under varying misconfiguration conditions.

### 4.3.2. Synthetic PAN Input

OSPOS, in its default configuration, does not process or store Primary Account Numbers (PANs), particularly within the controlled environment used in this study. However, SME deployments of web-based point-of-sale systems may, through misconfiguration or poor implementation practices, handle sensitive cardholder data in an insecure manner.

To simulate this behaviour in a controlled and ethical way, the existing 'Comments' field within the Sales interface was repurposed to accept synthetic PAN input, as shown in Figure 3. This approach enables the controlled injection of cardholder-like data into the application workflow, allowing analysis of its transmission, processing, storage and persistence across system components without modifying core payment functionality. '4111 1111 1111 1111' is a widely recognised Visa test card number, commonly used in payment gateway sandbox environments (PayPal, No Date). It was used in this test to simulate PAN data in an ethical way. The PAN alone constitutes cardholder data and must be protected accordingly, hence its use in this test.

## 4.4. Baseline Configuration

The screenshot displays a transaction summary with the following details:

| Delete | Type       | Amount  |
|--------|------------|---------|
|        | Debit Card | \$10.00 |

Below the summary, there are two buttons: a grey 'Suspend' button with a menu icon and a red 'Cancel' button with a close icon. At the bottom, a label 'Card Number (PAN)' is positioned above a text input field containing the test card number '4111 1111 1111 1111'. A green 'Complete' button with a checkmark is located at the top right of the interface.

Figure 3 – 'Comments' Field Repurposed for PAN Input

A secure baseline was designed to demonstrate how sensitive and transactional data should not be accessible in the various layers of the POS environment. This baseline would then also allow for 'misconfigurations', allowing the examination and demonstration of what specific misconfigurations cause the persistence of sensitive data, and the specific source for that data.

The OSPOS environment was secured in several ways, such as forcing HTTPS (encrypting transmitted data), this was achieved by creating self-signed certificates solely for the purposes of testing. Debug Mode was also disabled, and log information was limited, these were achieved through modifying the OSPOS and PHP config files. Encryption at rest was implemented by encrypting the PAN input before storing in the database. Other configuration hardening measures were taken, too, all of which are available in Appendix B – Empirical Misconfiguration Study and Results

## 4.5. Misconfiguration Scenario Selection

Five misconfiguration scenarios were selected for this experiment, covering a range of security risks, data remanence sources, and implications. Debug Mode and Verbose Logging were combined into one scenario, as they have similar misconfiguration implementations and data remanence sources, both demonstrating the security risks with logs containing excessive data. Insecure transmission was chosen as it demonstrates the security risks across the network layer and has a unique data remanence source. Session Storage Misuse was implemented to demonstrate security risks and data remanence on the client's side, which not only makes misconfiguration identification harder, but is also the side exposed to more users (both non-technical and malicious). Lack of encryption at rest was chosen to demonstrate the database security risks, which was then followed by improper data invalidation, which has a unique data remanence source involving data carving, demonstrating the common misconception that 'deleted' data doesn't necessarily mean 'erased' data.

## 4.6. Data Collection and Examination Procedure

Following the implementation of the secure baseline, three identical transactions were processed in the OSPOS environment using the synthetic PAN value. After baseline testing, VM snapshots were restored and each misconfiguration scenario was implemented individually. The same transaction process was repeated, and the expected artefact locations were examined using relevant methods, including file system analysis, SQL queries, log inspection, browser developer tools, and Wireshark packet captures. This ensured that the impact of each misconfiguration could be isolated and compared against the secure baseline.

# Chapter 5. Empirical Validation

## 5.1. Introduction

This chapter presents the empirical validation conducted within the controlled OSPOS environment. Following the foundation of the secure baseline, a series of ‘misconfiguration’ scenarios were implemented independently to assess whether sensitive transactional data could be recovered from expected artefact locations. The results from each scenario were then compared against the secure baseline to determine the impact of each misconfiguration on data persistence. The full, detailed test is available in Appendix B – Empirical Misconfiguration Study and Results

## 5.2. Test Environment

The testbed used a low-resource Ubuntu Server Virtual Machine (VM) hosting OSPOS 3.4.1 and a Windows 10 Thin-Client VM accessing the application through the Chrome browser.

The OSPOS sales page, as seen from the Thin-Client, can be seen in Figure 4. A receipt page was presented after each successful transaction, a sample of which is shown in Figure 5.

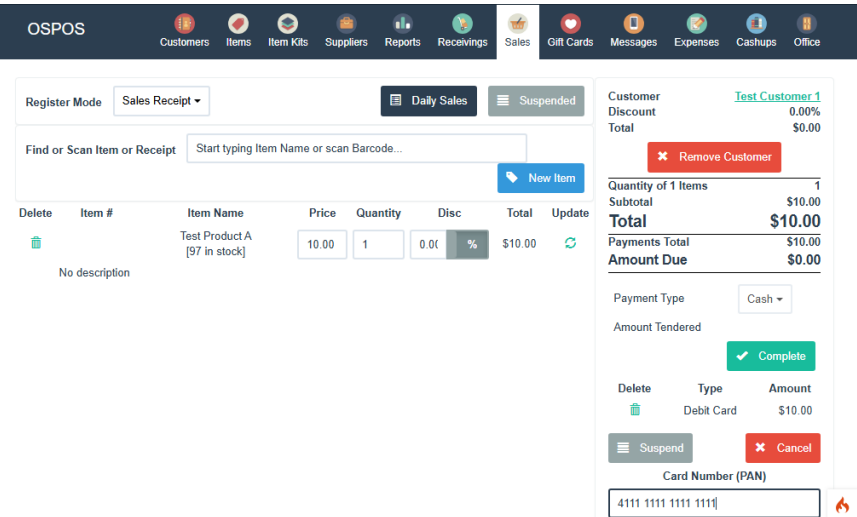


Figure 4 – OSPOS Sales Page

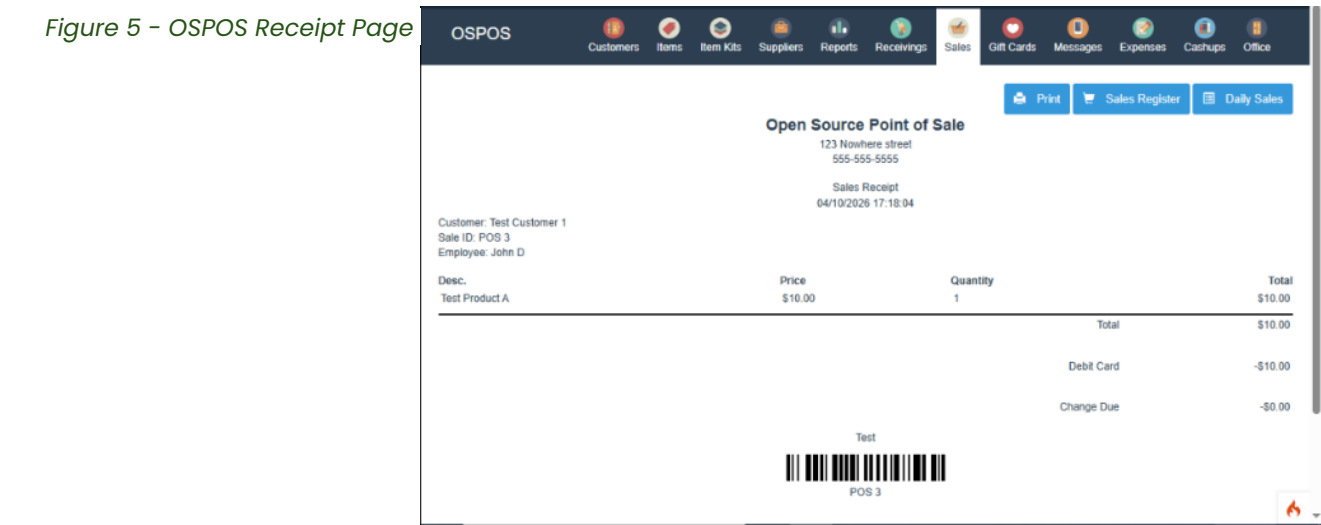


Figure 5 – OSPOS Receipt Page

### 5.3. Secure Baseline

Following the test transactions, client-side, server-side, and network artefact locations were examined, including database storage, logs, browser storage, temporary files, backups, and network packets.

As expected, these source locations did not contain PAN data, indicating that the secure baseline sufficiently secured sensitive data through minimal storage or encryption where necessary. This validation of the secure baseline is necessary for ensuring the following misconfiguration scenarios produced accurate and repeatable results.

### 5.4. Scenarios

The following scenarios were performed independently, with snapshot reversions between each test. Following the secure baseline test, encryption at rest was disabled for scenarios involving direct storage recovery and network transmission analysis. This ensured that plaintext PAN values could be identified within the database artefacts and captured packets, allowing the effect of each misconfiguration to be observed clearly.

#### 5.4.1. Scenario 1 – Debug Mode & Verbose Logging

Debug Mode was enabled in the OSPOS environment configuration file and verbose logging was implemented by increasing the logging threshold. Simulating a deployment in which development and testing settings were not reverted prior to production use.

The expected data remanence source locations included the application and webserver log directories, as these are the locations in log files are present. In a secure deployment, these logs would only contain essential information, providing an overview with no explicit data. Temporary files were also analysed, although presence was not expected.

Upon examination, log inspection took place, and the only data remanence source was the application logs, of which there were six, all containing the full PAN, indicating two log files per transaction. This was expected as these files are typically associated with debug mode and the logging threshold was high.

Interestingly, the webserver logs did not contain the PAN, which indicates that application-specific configuration does not influence the webserver. Furthermore, as expected, the PAN was not present in the temporary files.

These results demonstrate the importance of ensuring logs are kept to minimal, overview-like information, and the significance Debug Mode has in production environments.

The comparison between the secure baseline and misconfigured environment outputs are available in Figure 6.

| Secure Baseline Output                                                                                                | Misconfigured Implementation Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>posadmin@ubuntu:~/Desktop\$ sudo grep -ro "4111 1111 1111 1111" /var/www/html/ posadmin@ubuntu:~/Desktop\$</pre> | <pre>posadmin@ubuntu:~/Desktop\$ sudo grep -ro "4111 1111 1111 1111" /var/www/html/writable/debugbar debugbar_1775908133.926641.json:4111 1111 1111 1111 debugbar_1775908133.926641.json:4111 1111 1111 1111 debugbar_1775908136.273451.json:4111 1111 1111 1111 debugbar_1775908136.273451.json:4111 1111 1111 1111 debugbar_1775908136.273451.json:4111 1111 1111 1111 debugbar_1775908136.273451.json:4111 1111 1111 1111 debugbar_1775908136.273451.json:4111 1111 1111 1111 posadmin@ubuntu:~/Desktop\$</pre> |

Figure 6 – Verbose Logging Artefact Comparison

### 5.4.2. Scenario 2 – Insecure Transport

Insecure transport was implemented by allowing HTTP connections and, for this test, HTTPS was disabled entirely. This simulated a non-technical system administrator implementing TLS but failing to redirect HTTP traffic to HTTPS.

The expected data remanence source was network packets, as packets are the pieces of data that are transmitted across the network.

Upon examination, Wireshark packet capturing took place during the test transactions. These packets were analysed and the PAN was present in plaintext. In the secure baseline test, these packets were encrypted and the PAN would not be visible or recoverable without the decryption key.

These results illustrate the importance of encrypting transmitted data, as the network layer is susceptible to interception. Furthermore, plaintext traffic may also be retained within monitoring and logging systems, such as IDS or SIEM systems, increasing the persistence and exposure of sensitive data beyond the initial transmission itself.

The comparison between the secure baseline and misconfigured environment outputs are available in Figure 7.

| Secure Baseline Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Misconfigured Implementation Output                                                                                                                                                                                                                           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| <pre>POST /sales/complete HTTP/1.1 Host: 192.168.231.129 Connection: keep-alive Content-Length: 37 Cache-Control: max-age=0 Origin: http://192.168.231.129 Content-Type: application/x-www-form-urlencoded Upgrade-Insecure-Requests: 1 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7 Referer: http://192.168.231.129/sales/addPayment Accept-Encoding: gzip, deflate Accept-Language: en-US,en;q=0.9 Cookie: debug-bar-state=minimized; ospos_session=08cf85653a80aa0f6cca8c2d881d3258  honeypot=&amp;comment=4111+1111+1111+1111</pre> | <table><tr><th>Source</th><th>Destination</th><th>Protocol</th><th>...</th></tr><tr><td>192.168.231.130</td><td>192.168.231.129</td><td>TCP</td><td>...0...K...W...e...U...h...r...x...?...5...</td></tr><tr><td>192.168.231.129</td><td>192.168.231.130</td><td>TCP</td><td>...+.../...0...+...+.../...0...+...+.../...</td></tr><tr><td>192.168.231.130</td><td>192.168.231.129</td><td>TCP</td><td>...</td></tr><tr><td>192.168.231.130</td><td>192.168.231.129</td><td>TCP</td><td>...</td></tr><tr><td>192.168.231.130</td><td>192.168.231.129</td><td>TLSv1.3</td><td>...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+...+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Figure 7 – Insecure Transport Artefact Comparison

### 5.4.3. Scenario 3 – Client-side Session Storage Misuse

Client-side Session Storage Misuse was implemented by adding code to the sales register page that locally stores the PAN, simulating a configuration where convenience is prioritised over security by retaining transaction data locally within the browser.

The expected data remanence source was within the web browser's local/session storage, as the browser would be used to access the webapp and send data input to it.

Upon examination, developer tools were used and the local storage data was investigated in the receipt page of each transaction. The PAN was visible in all instances, alongside timestamps for the logging mechanisms.

This result demonstrates that security must have priority over convenience and client-side data exposure must be controlled to ensure optimal security.

The comparison between the secure baseline and misconfigured environment outputs are available in Figure 8.

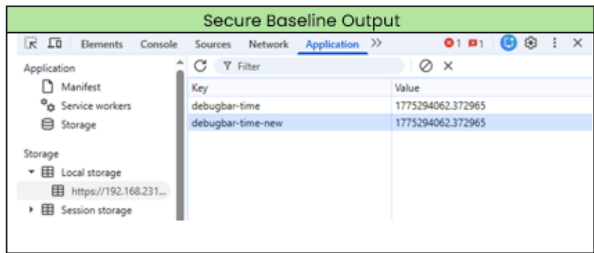
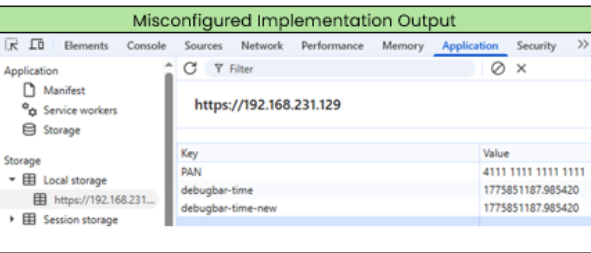
| Secure Baseline Output                                                            | Misconfigured Implementation Output                                                |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |

Figure 8 – Session Storage Misuse Artefact Comparison

#### 5.4.4. Scenario 4 – Lack of Encryption at Rest

To implement the Lack of Encryption at Rest misconfiguration, the PAN encryption step was removed. This simulates an insecure data storage environment that holds sensitive data in plaintext.

The expected source of data remanence was within the database, as they are designed to store a variety of data, such as employee data, customer data, login data and, in this case, transactional data.

Upon examination, SQL queries were used to retrieve data from within the database. During investigation, there were three rows containing the PAN – one for each transaction. Furthermore, file system analysis was performed to retrieve the same PAN data from the MySQL InnoDB table file, in both string and hex values, indicating that direct database authorisation is not necessarily required to view this data.

This result clearly shows the importance of encrypting data at rest, particularly sensitive data, as access to the device's file system (though, in most cases, with administrative privileges) may be enough to access this data.

The comparison between the secure baseline and misconfigured environment outputs are available in Figure 9.

| Secure Baseline Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Misconfigured Implementation Output                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>mysql&gt; SELECT sale_id, comment FROM ospos_sales ORDER BY sale_id DESC LIMIT 5; +-----+-----+   sale_id   comment   +-----+-----+   5   4tL3GoP9youpuZ/VamTF7D06Nz8ROT8vaxdMRHJEVdKL25rV1FYmo4ZHd4TndqbkgwRHV5bTVoV29YWT0=     4   49X4Laz1TGZ74x61CJDz4z06cDFLRku2aDBXMeSwT0LkY1Rnd05WejNZY2VNMzc2T25LRmFR00H0W/hrcz0=     3   57Z8IM/vvK+zuuysJngIKzo6aD330DFkNytRTGFUV1Iz2F31WjA3TX0Den9vdmpha0n0W6tTb0VDRHdXMD0=   +-----+-----+ 3 rows in set (0.01 sec)</pre> <pre>posadmin@ubuntuuser:~\$ sudo strings /var/lib/mysql/ospos/ospos_sales.ibd   grep 4111 posadmin@ubuntuuser:~\$</pre> | <pre>mysql&gt; SELECT sale_id, comment FROM ospos_sales ORDER BY sale_id DESC LIMIT 5; +-----+-----+   sale_id   comment   +-----+-----+   5   4111 1111 1111 1111     4   4111 1111 1111 1111     3   4111 1111 1111 1111   +-----+-----+ 3 rows in set (0.00 sec)</pre> <pre>posadmin@ubuntuuser:~\$ sudo strings /var/lib/mysql/ospos/ospos_sales.ibd   grep 4111 4111 1111 1111 1111 4111 1111 1111 1111 4111 1111 1111 1111 posadmin@ubuntuuser:~\$</pre> |

Figure 9 – Lack of Encryption Artefact Comparison

#### 5.4.5. Scenario 5 – Improper Invalidation

Improper invalidation is the only scenario in this test that does not require a direct 'misconfiguration'. Instead, the implementation involves accessing the MySQL database and using SQL queries to delete a row of PAN data from the 'sales' table. After doing so, further SQL queries were used to view the table, showing only two transactions and PANs, rather than three.

The expected source of data remanence was within the underlying storage, such as database files or the system's unallocated space.

Upon examination, file system analysis was performed to investigate the database files. Data recovery was inconsistent, but possible in most cases, in which the PAN data would *still* be within the InnoDB table file while not present in the database structure, even after system restart.



While row-by-row deletion did not immediately overwrite the underlying InnoDB table file, subsequent operations – such as ‘TRUNCATE’ or ‘OPTIMISE’, which rebuild the table – triggered physical updates that completely erased the PAN data. This observation indicates that data remanence exists after logical deletion, but is influenced by subsequent database activity and behaviour.

These results demonstrate how data can remain even after deletion, contrary to a common misconception that ‘deleted’ means ‘securely erased’, making this a security risk even if the ‘deleted’ data was encrypted.

shows the database and storage analysis outputs before and after deleting a row.

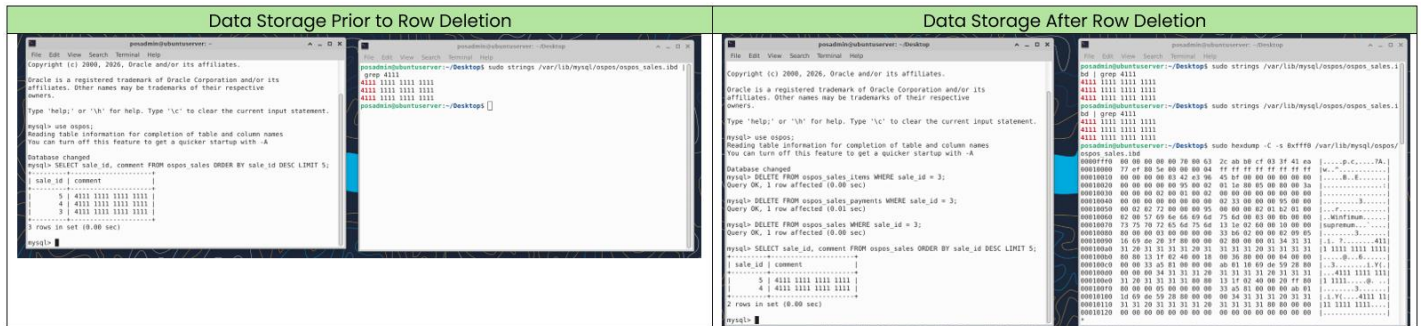


Figure 10 – Improper Invalidation Artefact Comparison

## 5.5. Chapter Summary

This experiment illustrates how different misconfigurations can produce different artefacts in different locations and, by implementing each misconfiguration one-at-a-time, it is clear what specific misconfigurations cause the recoverability of sensitive data.

Furthermore, the literature findings and analytical framework enabled the accurate prediction of remanence source locations, as the results broadly matched the expected outcomes, also verifying the accuracy of the secure baseline test.

## Chapter 6. Discussion

### 6.1. Introduction

This chapter synthesises the literature review and empirical validation findings to answer the research questions with accuracy and verifiability, as well as evaluate the limitations and consider the implications for SMEs.

### 6.2. Answering Research Questions

#### 6.2.1. RQ1 – Prominent Misconfiguration Risks

*What misconfiguration-related security risks are most prominent in web-based POS deployments used by SMEs?*

Common root causes of misconfigurations include human error, lack of technical expertise, insecure default settings, poor lifecycle management, and the prioritisation of convenience over security. In SME environments, these factors may result in applications being deployed with default configurations, excessive logging, weak session handling, and inadequate data retention controls.

The OWASP Top 10 (2025b) categorises common web application vulnerabilities, with Security Misconfiguration in second-place position on the list, rising from fifth place in the 2021 edition, indicating a growing concern around the impact of configuration weaknesses in web applications. Further vulnerabilities included Cryptographic Failures in fourth place, and Security Logging and Alerting Failures in ninth place. PCI DSS (2024) provides compliance requirements regarding sensitive transactional data, alongside comprehensive storage requirements, requiring such data to be non-existent, or encrypted where storage is necessary.

The empirical validation simulated various selected misconfiguration scenarios impacting different areas of a controlled, web-based POS environment. Each scenario caused data persistence in different source locations, where sensitive data artefacts were recovered, primarily through storage analysis. Logging misconfigurations caused sensitive transactional data to be written into log files during normal operational use, insecure transport caused sensitive transactional data to be accessible during network transmission, where those packets may be stored by intrusion detection systems and network monitoring tools, client-side storage misuse caused sensitive transactional data to be written in plaintext on the device accessing the webapp, unencrypted database tables caused sensitive transactional data to be retrievable without direct access to the database, and improper invalidation allowed sensitive transactional data to be recovered after ‘deletion’.

Combining these sources of information, cryptography emerged as a recurring factor across multiple scenarios. While lack of encryption at rest and insecure transport are, themselves, prominent risks, encryption also reduces the severity of other misconfigurations. For example, verbose logging and poor invalidation may still result in residual artefacts, but encrypted data has significantly less detrimental impact upon recovery.

Overall, the most prominent misconfiguration-related security risks in SME POS environments are verbose logging, insecure transport, poor invalidation, session mismanagement, and inadequate encryption. Of these, lack of cryptographic protection appears to be particularly significant, as effective encryption can reduce the impact of several other misconfigurations where sensitive data remains recoverable, further supporting SME compliance.

### 6.2.2. RQ2 – Framework Coverage and SME Gaps

*How do existing guidance and compliance frameworks, including Cyber Essentials, OWASP, and PCI DSS address these risks and where do practical gaps remain for SMEs?*

Across NCSC Cyber Essentials, OWASP Foundation Guidance, and PCI SSC standards, the core misconfiguration risks – including verbose logging, insecure transport, poor invalidation, session mismanagement, and lack of encryption at rest – are consistently recognised at a conceptual level.

PCI DSS addresses these risks very clearly, providing best practices and detailed requirements for compliance, however, it focuses on these rather than system-specific implementation details. In contrast, OWASP's frameworks offer stronger application-level guidance, particularly through ASVS, but remain fragmented, being distributed across multiple resources, and assume a level of technical expertise that may not be present in SME environments. Cyber Essentials provides an accessible baseline for cyber hygiene, but lacks the depth required to address application-specific configuration risks.

Collectively, these frameworks define what secure practice *should* look like, but they do not sufficiently bridge the gap between high-level requirements and real-world SME implementation. This creates a practical disconnect, where SMEs are aware of security expectations, but lack the expertise, resources, or contextual guidance to implement them effectively. As a result, security controls may be partially implemented, misconfigured, or applied inconsistently, directly increasing the likelihood of misconfiguration-related vulnerabilities. This limitation is especially critical in web-based POS environments, where incorrect implementation of otherwise well-defined controls can directly lead to the persistence and recoverability of sensitive transactional data across system components, as demonstrated in the empirical validation, highlighting the practical consequences of this gap in SME environments.

### 6.2.3. RQ3 – Empirical Validation Outcome

*To what extent do selected misconfiguration scenarios lead to recoverable residual artefacts in a controlled open-source POS test environment?*

The empirical validation was conducted within a controlled point-of-sale environment, beginning with a verified secure baseline configuration. Selected misconfiguration scenarios – including debug mode and verbose logging, insecure transport, client-side session storage misuse, lack of encryption at rest, and poor invalidation – were then implemented, and predicted artefact locations were examined for residual sensitive data.

Across all test scenarios, misconfigurations resulted in the recovery of sensitive transactional data from at least one source across the client-side, server-side, or network layers, demonstrating that these risks are clearly present within the controlled environment. Logging misconfigurations led to the persistence of PAN data within application logs, while insecure transport exposed plaintext data during network transmission, increasing the risk of interception and unintended retention within monitoring systems. Client-side storage misuse resulted in sensitive data being retained within browser storage, increasing exposure to the user, and lack of encryption at rest allowed sensitive data to be recovered directly from server-side storage, even without authorised database access.

Improper invalidation demonstrated that data considered to be deleted may remain recoverable within underlying storage. However, recovery was inconsistent, an important finding that suggests that data remanence is influenced not only by misconfigurations, but also by system behaviour and usage. This variability highlights the difficulty SMEs may face in reliably ensuring secure data deletion – an important requirement for PCI DSS compliance – and performing reliable audits.

These findings align with the literature and established frameworks, particularly PCI SSC guidance, which emphasises encryption, secure deletion, and controlled data handling. Overall, the results demonstrate that misconfiguration-related risks can lead to recoverable residual artefacts to a significant extent, and that even relatively minor configuration errors can result in sensitive data being retained across multiple system layers, increasing both exposure risk and forensic recoverability. These findings reinforce that misconfiguration represents a practical and high-impact risk in SME POS environments, particularly where encryption and data lifecycle controls are insufficiently implemented.

### 6.3. Ethical and Limitation Considerations

The empirical validation was limited to a single open-source POS platform with repurposed application fields, which may not fully represent the behaviour of other systems, particularly proprietary deployments with transaction-handling capabilities. The use of synthetic PAN data, while necessary for ethical reasons, also may not fully reflect real-world operation.

Furthermore, this study focused on a small set of selected misconfiguration scenarios for analysis, and did not examine longer-term persistence, memory artefacts, or backup systems. As a result, while the findings are analytically valuable, they are not fully generalisable and should be interpreted as indicative rather than definitive.

### 6.4. Implications

These findings highlight the need for more practical, implementation-focused security guidance for SMEs, as existing frameworks do not sufficiently support the translation of high-level requirements into real-world configurations. In particular, greater emphasis should be placed on data lifecycle management, rather than breach prevention alone, as sensitive data may persist even in the absence of active compromise. The results also suggest that SMEs may underestimate the risks associated with data remanence, particularly where ‘deleted’ data remains recoverable, creating both security and compliance challenges that are not adequately addressed by current guidance.

## Chapter 7. Conclusion

### 7.1. Summary of Findings

This study examined the impact of misconfiguration-related security risks in web-based POS deployments used by SMEs, with particular focus on how such misconfigurations may lead to the persistence and recoverability of sensitive transactional data. Through a combination of secondary literature analysis and empirical validation, the research identified key misconfiguration risks, evaluated the effectiveness of existing security frameworks, and assessed the extent to which these risks materialise in practice. The findings demonstrate that misconfiguration is not only a common security issue, but also a significant contributor to data remanence, where sensitive data may persist across multiple system layers, even without the existence of active exploitation.

### 7.2. Response to Research Questions

The most prominent misconfiguration-related security risks in SME POS environments include verbose logging, insecure transport, improper data invalidation, poor session handling, and inadequate encryption. These risks are particularly significant due to SME-specific factors such as limited technical expertise, reliance on default configurations, and insufficient lifecycle management practices. Among these, the lack of cryptographic protection is particularly critical, as effective encryption can reduce the impact of multiple other misconfigurations.

Existing frameworks, including Cyber Essentials, OWASP, and PCI DSS, collectively define secure practices at a conceptual level, but do not adequately support SMEs in translating these requirements into practical implementation. While PCI DSS provides detailed compliance requirements, OWASP offers web application-level guidance, and Cyber Essentials provides accessible baseline security, all three frameworks lack sufficient implementation guidance, particularly for SMEs. This creates a practical gap where security controls may be incorrectly implemented, increasing the likelihood of misconfiguration-related vulnerabilities.

The empirical validation demonstrated that selected misconfiguration scenarios consistently resulted in recoverable residual artefacts across client-side, server-side, and network layers. All tested scenarios produced observable data remanence, confirming that misconfiguration risks manifest to a significant extent within the controlled POS environment. While some variability was identified, particularly in improper data invalidation, the results confirm that even minor configuration errors can lead to the persistence and recoverability of sensitive data.

### 7.3. Contributions of the Study

This study contributes to the existing body of knowledge by providing a structured analysis of how common SME-relevant misconfigurations lead to both security vulnerabilities and forensic data persistence. By developing a conceptual framework linking misconfiguration types to residual artefact locations, followed by a validation of these relationships through empirical testing, this research offers a more integrated understanding of misconfiguration as a security risk. Additionally, it highlights the practical limitations of existing security frameworks in SME contexts, particularly in bridging the gap between theoretical guidance and real-world implementation.

## 7.4. Recommendations for Future Research

Future work could extend this study by analysing multiple POS platforms, including proprietary systems with transaction-handling capabilities, and examining additional artefact sources, such as short-term volatile memory, long-term backup systems, and forensic data-carving. Long-term persistence analysis and real-world SME deployments would further demonstrate how these risks manifest over time.

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# Appendices

## Appendix A – Critical Appraisal

This project developed from an initial interest in mobile point-of-sale (POS) terminals and the security of commercially available payment devices that can be configured and used by consumers. However, after facing research and ethical limitations, the project shifted from consumer environments to small and medium enterprise (SME) environments, as the scale of these POS systems provided larger vulnerability vectors for analysis. During the research process, the scope was refined to focus specifically on misconfiguration, data persistence, and recoverable artefacts rather than actively exploitable vulnerabilities to contribute to a little-explored literature gap.

The literature review was one of the most challenging parts of the project as the topic spans across several areas, including SME cyber security practices and constraints, POS systems, compliance frameworks, web application security, and digital forensics. While relevant sources existed from each area, few studies linked them directly, which influenced the development of the analytical framework, which became an important way of synthesising the literature.

The controlled testbed provided a useful way to validate the theoretical framework. Using OSPOS allowed the system to be modified ethically and repeatably, while synthetic PAN input avoided the use of real payment data. The use of virtual machine snapshots helped isolate each misconfiguration scenario and improved the reliability of the results. However, the test environment was limited to one platform and a small number of selected scenarios, meaning the findings should be interpreted as indicative rather than fully generalisable.

A key strength of the work is that it connects misconfiguration to forensic data persistence, rather than treating misconfiguration as solely an exploitation risk. The analytical framework also provides a structured way to link misconfiguration types, artefact locations, and security consequences. The empirical chapter strengthens the work by demonstrating that the theoretical risks can produce recoverable artefacts in practice.

Testing multiple POS platforms and analysing additional artefacts scenarios, such as memory and backup systems, would strengthen the empirical validation. Conducting longer-term persistence analyses and using forensic techniques, such as data-carving, would also improve the rigour of this study.

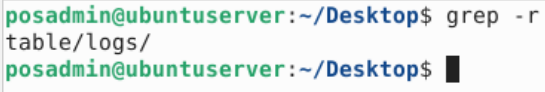
This project significantly improved my understanding of how academic research, standards analysis, and empirical testing can be combined within a cyber security study. It has developed my skills in literature synthesis, experimental design, and critical evaluation. The most prominent lesson I have learned is that a strong project isn't just about producing technical results, but about clearly explaining *why* those results matter.

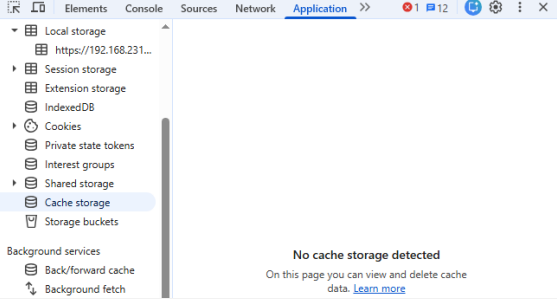
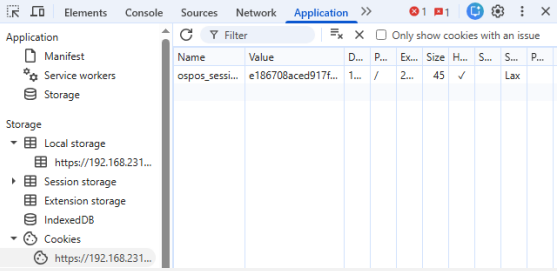
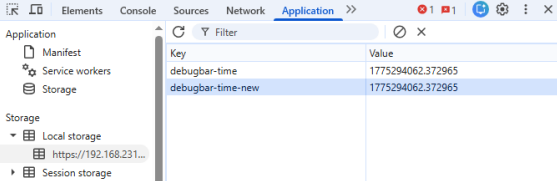


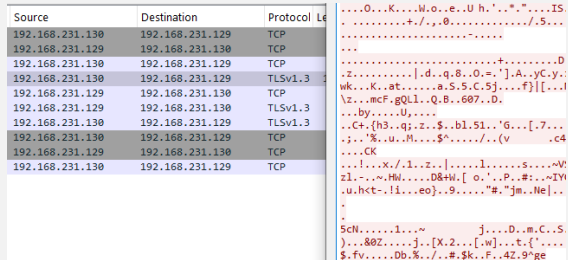
## Appendix B – Empirical Misconfiguration Study and Results

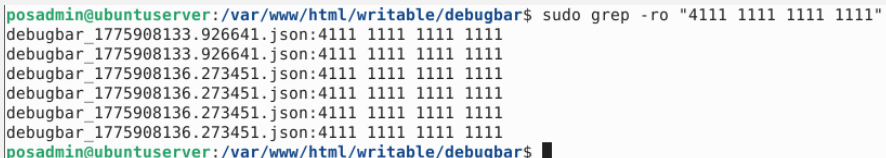
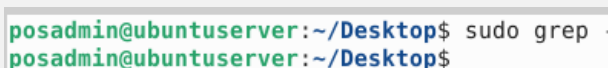
| Secure Baseline Development                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Virtual Machines                                         | Linux Ubuntu Server 24.04.3 LTS, 2 Processors, 2GB RAM, Snapshot Taken<br>Windows 10 22H2, 2 Processors, 2GB RAM, Snapshot Taken                                                                                                                                                                                                                                                                                                                             |
| LAMP Hardening                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Action (Linux)                                           | Command(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                          | In-Text Change(s)                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Update OS                                                | \$ sudo apt update && sudo apt upgrade -y                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Apache Hardening                                         | \$ sudo nano /etc/apache2/conf-available/security.conf                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                          | ServerTokens Prod<br>ServerSignature Off<br>TraceEnable Off                                                                                                                                                                                                                                                                                                                                                                                                  |
| Disable Directory Listing                                | \$ sudo nano /etc/apache2/apache2.conf                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                          | <Directory /var/www/><br>Options -Indexes FollowSymLinks<br>AllowOverride All<br>Require all granted<br></Directory>                                                                                                                                                                                                                                                                                                                                         |
| Enable HTTPS                                             | \$ sudo a2enmod ssl<br>\$ sudo a2ensite default-ssl<br>\$ sudo systemctl reload apache2<br>\$ sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/ssl/private/ospos.key -out /etc/ssl/certs/ospos.crt<br>\$ sudo nano /etc/apache2/sites-available/default-ssl.conf                                                                                                                                                                        |
|                                                          | SSLCertificateFile /etc/ssl/certs/ospos.crt<br>SSLCertificateKeyFile /etc/ssl/private/ospos.key                                                                                                                                                                                                                                                                                                                                                              |
| PHP Hardening                                            | \$ sudo nano /etc/php/*/apache2/php.ini                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                          | display_errors = Off<br>display_startup_errors = Off<br>log_errors = On<br>error_log = /var/log/php_errors.log<br><br>expose_php = Off<br><br>session.cookie_httponly = 1<br>session.cookie_secure = 1                                                                                                                                                                                                                                                       |
| MySQL Hardening                                          | \$ sudo mysql_secure_installation                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                          | ^ Remove anonymous users? Y<br>^ Disallow root login remotely? Y<br>^ Remove test database and access to it? Y<br>^ Reload privilege tables now? Y                                                                                                                                                                                                                                                                                                           |
| Logging                                                  | \$ sudo nano /var/www/html/app/Config/Logger.php                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                          | public \$threshold = 0;                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Encrypting PAN Input<br>(Disabled during scenario tests) | \$ sudo nano /var/www/html/app/Controllers/Sales.php                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                          | Remove:<br>\$data['comments'] = \$this->sale_lib->get_comment();<br><br>Add:<br>\$plain_pan = \$this->sale_lib->get_comment();<br>\$encryption_key = 'panencrypt123';<br>\$cipher = 'AES-256-CBC';<br>\$iv_length = openssl_cipher_iv_length(\$cipher);<br>\$iv = random_bytes(\$iv_length);<br>\$encrypted_pan = openssl_encrypt(\$plain_pan, \$cipher,<br>\$encryption_key, 0, \$iv);<br>\$data['comments'] = base64_encode(\$iv . ':' . \$encrypted_pan); |

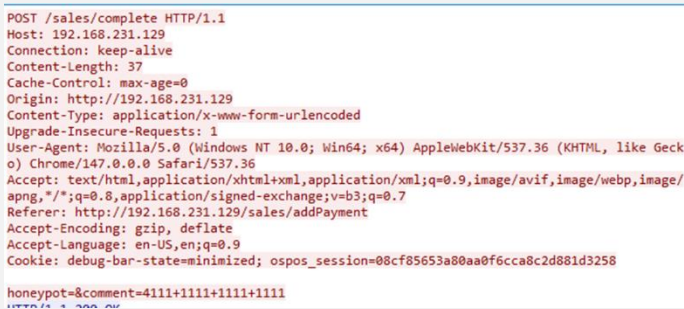
| Methodology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Three transactions on the Thin-Client VM, all the same (this ensures no false negatives caused by failed transactions):</p> <ul style="list-style-type: none"> <li>Item: 'Test Product A' – £10.00</li> <li>Quantity: 1</li> <li>Customer 'Test Customer 1'</li> <li>Payment Type: 'Debit Card'</li> <li>Card Number (PAN)*: 4111 1111 1111 1111</li> </ul> <p>* Previously 'Comments' field</p> <p>Each predicted data remanence source location will be analysed.</p> <p>After the Secure Baseline test, each VM will be reverted to the 'Secure Baseline' snapshots. Then one misconfiguration will be implemented and changes will be documented. Relevant predicted data remanence source locations will be analysed and findings will be recorded.</p> <p>After each misconfiguration scenario, each VM will be reverted to the 'Secure Baseline' snapshots before proceeding to the next scenario.</p> |

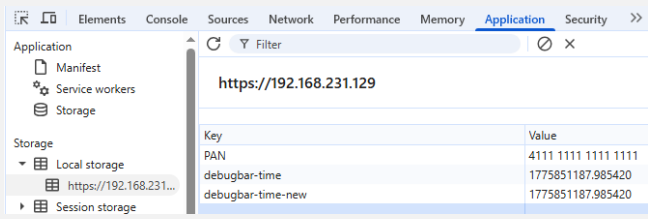
| Secure Baseline Test   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Database Storage       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Affected Layer         | Server-side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Expected Presence      | Present, Encrypted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Resulting Presence     | Present, Encrypted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Steps Taken            | <pre>\$ sudo mysql USE ospos; SELECT sale_id, comment FROM ospos_sales ORDER BY sale_id DESC LIMIT 5; \$ sudo strings /var/lib/mysql/ospos/ospos_sales.ibd   grep 4111</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Supporting Screenshots |  <pre>mysql&gt; SELECT sale_id, comment FROM ospos_sales ORDER BY sale_id DESC LIMIT 5; +-----+-----+   sale_id   comment   +-----+-----+   5         4tL3GoP8yowpuZ/VamTF7Do6NzBR0TBvaXdMRHJEwVdKL25rV1FYWno4ZHd4TndqbkgrRHV5bTVoV29YwT0=     4         49X4laz1TGZ74x61CJDz4zo6cDFLRkw2aDBXWm5wT0lkY1Rnd05WejN2Y2VnKzc2T25LRmFrMHHHwVhrcz0=     3         57ZBIM/vvK+zuuysJngIKzo6aDJ3bDFkNytRTGFUV1IzZFJ1WjA3TXRDem9vdmphaDhDwGtTbDVRHdxMD0=   +-----+-----+ 3 rows in set (0.01 sec)</pre> <pre>posadmin@ubuntuuserver:~\$ sudo strings /var/lib/mysql/ospos/ospos_sales.ibd   grep 4111 posadmin@ubuntuuserver:~\$</pre> |
| Application Logs       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Affected Layer         | Server-side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Expected Presence      | Not Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Resulting Presence     | Not Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Steps Taken            | <pre>\$ sudo grep -ro "4111 1111 1111 1111" /var/www/html/</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Supporting Screenshots |  <pre>posadmin@ubuntuuserver:~/Desktop\$ grep -r table/logs/ posadmin@ubuntuuserver:~/Desktop\$</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Web Server Logs        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Affected Layer         | Server-side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Expected Presence      | Not Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Resulting Presence     | Not Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Steps Taken            | <pre>\$ sudo grep -ro "4111 1111 1111 1111" /var/log/apache2/</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Supporting Screenshots |  <pre>posadmin@ubuntuuserver:/var/www/html\$ sudo grep -ro "41 posadmin@ubuntuuserver:/var/www/html\$</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

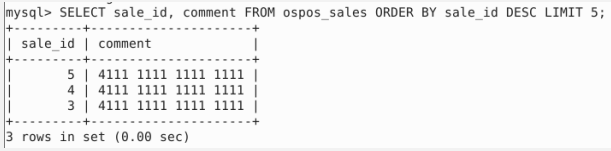
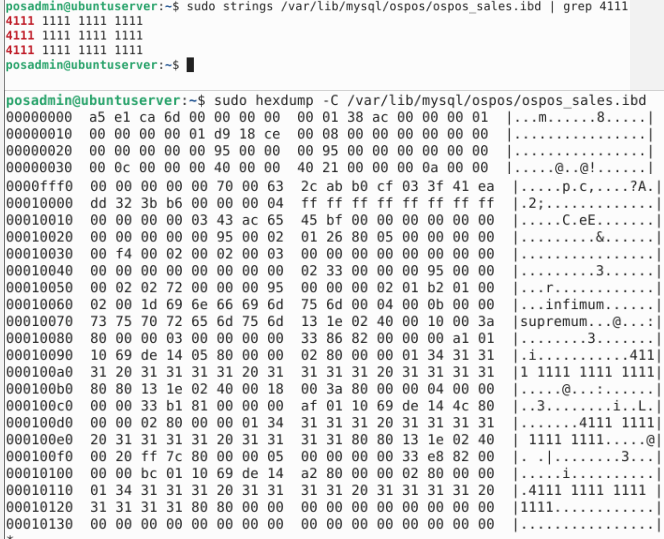
| Browser Cache           |                                                                                                                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Affected Layer          | Client-side                                                                                                                                                                                                                           |
| Expected Presence       | Not Present                                                                                                                                                                                                                           |
| Resulting Presence      | Not Present                                                                                                                                                                                                                           |
| Steps Taken             | Open DevTools (FN+F12)<br>Check Network tab, Sources tab, Application Tab (Cache Storage)                                                                                                                                             |
| Supporting Screenshots  |                                                                                                                                                     |
| Browser Cookies         |                                                                                                                                                                                                                                       |
| Affected Layer          | Client-side                                                                                                                                                                                                                           |
| Expected Presence       | Not Present                                                                                                                                                                                                                           |
| Resulting Presence      | Not Present                                                                                                                                                                                                                           |
| Steps Taken             | Open DevTools (FN+F12)<br>Check Application Tab (Cookies)                                                                                                                                                                             |
| Supporting Screenshots  |                                                                                                                                                    |
| Browser Session Storage |                                                                                                                                                                                                                                       |
| Affected Layer          | Client-side                                                                                                                                                                                                                           |
| Expected Presence       | Not Present                                                                                                                                                                                                                           |
| Resulting Presence      | Not Present                                                                                                                                                                                                                           |
| Steps Taken             | Open DevTools (FN+F12)<br>Check Application Tab (Local Storage, Session Storage, Shared Storage)                                                                                                                                      |
| Supporting Screenshots  |                                                                                                                                                   |
| Temporary Files         |                                                                                                                                                                                                                                       |
| Affected Layer          | Server-side                                                                                                                                                                                                                           |
| Expected Presence       | Not Present                                                                                                                                                                                                                           |
| Resulting Presence      | Not Present                                                                                                                                                                                                                           |
| Steps Taken             | \$ sudo grep -ro "4111 1111 1111 1111" /tmp/                                                                                                                                                                                          |
| Supporting Screenshots  |  |
| Backup Test             |                                                                                                                                                                                                                                       |

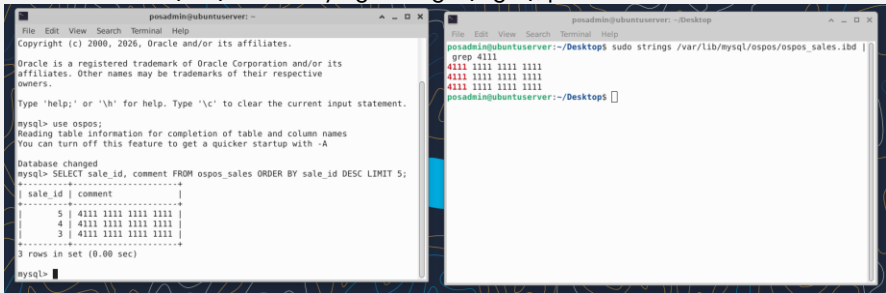
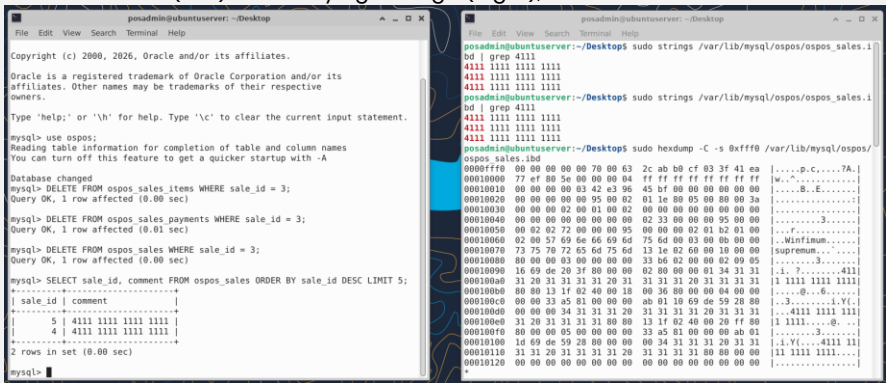
|                        |                                                                                    |
|------------------------|------------------------------------------------------------------------------------|
| Affected Layer         | Server-side                                                                        |
| Expected Presence      | Not Present (No backups made)                                                      |
| Resulting Presence     | Not Present (No backups made)                                                      |
| Steps Taken            | N/A                                                                                |
| Supporting Screenshots | N/A                                                                                |
| <b>Network Packets</b> |                                                                                    |
| Affected Layer         | Network Layer                                                                      |
| Expected Presence      | Present, Encrypted                                                                 |
| Resulting Presence     | Present, Encrypted                                                                 |
| Steps Taken            | Wireshark captured network packets during transaction transmission.                |
| Supporting Screenshots |  |

|                                                    |                                                                                                                                                                                                                                                                 |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Misconfiguration Scenario Tests</b>             |                                                                                                                                                                                                                                                                 |
| <b>Scenario 1 – Debug Mode and Verbose Logging</b> |                                                                                                                                                                                                                                                                 |
| Implementation Steps                               | <pre>\$ sudo nano /var/www/html/.env ^ CI_ENVIRONMENT = development \$ sudo nano /var/www/html/app/Config/Logger.php ^ public \$threshold = 9; \$ sudo systemctl restart apache2</pre>                                                                          |
| <b>Application Logs</b>                            |                                                                                                                                                                                                                                                                 |
| Affected Layer                                     | Server-side                                                                                                                                                                                                                                                     |
| Expected Presence                                  | Present                                                                                                                                                                                                                                                         |
| Resulting Presence                                 | Present                                                                                                                                                                                                                                                         |
| Steps Taken                                        | \$ sudo grep -ro "4111 1111 1111 1111" /var/www/html/                                                                                                                                                                                                           |
| Supporting Screenshots                             |  |
| <b>Web Server Logs</b>                             |                                                                                                                                                                                                                                                                 |
| Affected Layer                                     | Server-side                                                                                                                                                                                                                                                     |
| Expected Presence                                  | Present                                                                                                                                                                                                                                                         |
| Resulting Presence                                 | Not Present                                                                                                                                                                                                                                                     |
| Steps Taken                                        | \$ sudo grep -ro "4111 1111 1111 1111" /var/log/apache2/                                                                                                                                                                                                        |
| Supporting Screenshots                             |                                                        |
| <b>Temporary Files</b>                             |                                                                                                                                                                                                                                                                 |
| Affected Layer                                     | Server-side                                                                                                                                                                                                                                                     |
| Expected Presence                                  | Not Present                                                                                                                                                                                                                                                     |
| Resulting Presence                                 | Not Present                                                                                                                                                                                                                                                     |
| Steps Taken                                        | \$ sudo grep -ro "4111 1111 1111 1111" /tmp/                                                                                                                                                                                                                    |
| Supporting Screenshots                             |                                                                          |

| Scenario 2 – Insecure Transport |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation Steps            | <pre>\$ sudo a2dissite default-ssl~ \$ sudo a2dismod ssl \$ sudo systemctl reload apache2</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Network Packets                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Affected Layer                  | Network Layer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Expected Presence               | Present, Plaintext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Resulting Presence              | Present, Plaintext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Steps Taken                     | Wireshark packet capturing running throughout transactions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Supporting Screenshots          |  <pre>POST /sales/complete HTTP/1.1 Host: 192.168.231.129 Connection: keep-alive Content-Length: 37 Cache-Control: max-age=0 Origin: http://192.168.231.129 Content-Type: application/x-www-form-urlencoded Upgrade-Insecure-Requests: 1 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7 Referer: http://192.168.231.129/sales/addPayment Accept-Encoding: gzip, deflate Accept-Language: en-US,en;q=0.9 Cookie: debug-bar-state=minimized; ospos_session=08cf85653a80aa0f6cca8c2d881d3258  honeypot=&amp;comment=4111+1111+1111+1111</pre> |

| Scenario 3 – Client-side Session Storage Misuse |                                                                                                                                                                                                                                                                                                                                                                                       |     |       |     |                     |               |                   |                   |                   |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-----|---------------------|---------------|-------------------|-------------------|-------------------|
| Implementation Steps                            | <pre>\$ sudo nano /var/www/html/app/Views/sales/register.php  \$('#finish_sale_button').click(function() {     var pan = document.querySelector('#comment')?.value    '';     localStorage.setItem('PAN', pan);      \$('#buttons_form').attr('action', "&lt;?=" "\$controller_name/complete" "&gt;");     \$('#buttons_form').submit(); });  \$ sudo systemctl restart apache2</pre> |     |       |     |                     |               |                   |                   |                   |
| Browser Session Storage                         |                                                                                                                                                                                                                                                                                                                                                                                       |     |       |     |                     |               |                   |                   |                   |
| Affected Layer                                  | Client-side                                                                                                                                                                                                                                                                                                                                                                           |     |       |     |                     |               |                   |                   |                   |
| Expected Presence                               | Present                                                                                                                                                                                                                                                                                                                                                                               |     |       |     |                     |               |                   |                   |                   |
| Resulting Presence                              | Present                                                                                                                                                                                                                                                                                                                                                                               |     |       |     |                     |               |                   |                   |                   |
| Steps Taken                                     | On the '/sales/complete' page after a transaction, within the browser: DevTools (F12) > 'Application' Tab > 'Local Storage' Dropdown Pane                                                                                                                                                                                                                                             |     |       |     |                     |               |                   |                   |                   |
| Supporting Screenshots                          |  <table border="1"> <thead> <tr> <th>Key</th> <th>Value</th> </tr> </thead> <tbody> <tr> <td>PAN</td> <td>4111 1111 1111 1111</td> </tr> <tr> <td>debugbar-time</td> <td>1775851187.985420</td> </tr> <tr> <td>debugbar-time-new</td> <td>1775851187.985420</td> </tr> </tbody> </table>          | Key | Value | PAN | 4111 1111 1111 1111 | debugbar-time | 1775851187.985420 | debugbar-time-new | 1775851187.985420 |
| Key                                             | Value                                                                                                                                                                                                                                                                                                                                                                                 |     |       |     |                     |               |                   |                   |                   |
| PAN                                             | 4111 1111 1111 1111                                                                                                                                                                                                                                                                                                                                                                   |     |       |     |                     |               |                   |                   |                   |
| debugbar-time                                   | 1775851187.985420                                                                                                                                                                                                                                                                                                                                                                     |     |       |     |                     |               |                   |                   |                   |
| debugbar-time-new                               | 1775851187.985420                                                                                                                                                                                                                                                                                                                                                                     |     |       |     |                     |               |                   |                   |                   |

| Scenario 4 – Lack of Encryption at Rest |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation Steps                    | <p>Revert the PAN encryption mechanism back to its default configuration:</p> <pre>\$ sudo nano /var/www/html/app/Controllers/Sales.php</pre> <p>^ Remove:</p> <pre>// Take and Encrypt PAN   \$plain_pan = \$this-&gt;sale_lib-&gt;get_comment();  \$encryption_key = ''; \$cipher = 'AES-256-CBC'; \$iv_length = openssl_cipher_iv_length(\$cipher); \$iv = random_bytes(\$iv_length);  \$encrypted_pan = openssl_encrypt(\$plain_pan, \$cipher, \$encryption_key, 0, \$iv);  // Store IV and Ciphertext together \$data['comments'] = base64_encode(\$iv . ':' . \$encrypted_pan);</pre> <p>^ Add:</p> <pre>\$data['comments'] = \$this-&gt;sale_lib-&gt;get_comment();</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Database Storage                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Affected Layer                          | Server-side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected Presence                       | Present, Plaintext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Resulting Presence                      | Present, Plaintext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Steps Taken                             | <pre>\$ sudo mysql USE ospos; SELECT sale_id, comment FROM ospos_sales ORDER BY sale_id DESC LIMIT 5;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Supporting Screenshots                  |  <pre>mysql&gt; SELECT sale_id, comment FROM ospos_sales ORDER BY sale_id DESC LIMIT 5; +-----+-----+   sale_id   comment   +-----+-----+   5         4111 1111 1111 1111     4         4111 1111 1111 1111     3         4111 1111 1111 1111   +-----+-----+ 3 rows in set (0.00 sec)</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Underlying Storage Analysis             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Affected Layer                          | Server-side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Expected Presence                       | Unknown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Resulting Presence                      | Present, Plaintext                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Steps Taken                             | <pre>\$ sudo strings /var/lib/mysql/ospos/ospos_sales.ibd   grep 4111 \$ sudo hexdump -C /var/lib/mysql/ospos/ospos_sales.ibd</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Supporting Screenshots                  |  <pre>posadmin@ubuntu:~\$ sudo strings /var/lib/mysql/ospos/ospos_sales.ibd   grep 4111 4111 1111 1111 1111 4111 1111 1111 1111 4111 1111 1111 1111 posadmin@ubuntu:~\$ sudo hexdump -C /var/lib/mysql/ospos/ospos_sales.ibd 00000000 a5 e1 ca 6d 00 00 00 00 00 01 38 ac 00 00 00 01  ...m.....8....  00000010 00 00 00 00 01 d9 18 ce 00 08 00 00 00 00 00  .....  00000020 00 00 00 00 00 95 00 00 00 95 00 00 00 00 00  .....  00000030 00 0c 00 00 00 40 00 00 40 21 00 00 00 0a 00  ....@..@!.....  00000040 00 00 00 00 00 70 00 63 2c ab b0 cf 03 3f 41 ea  ....p.c....?A..  00000050 dd 32 3b b6 00 00 00 04 ff ff ff ff ff ff ff  .2;.....  00000060 00 00 00 00 03 43 ac 65 45 bf 00 00 00 00 00  ....C.eE.....  00000070 00 00 00 00 00 95 00 02 01 26 80 05 00 00 00  .....&amp;.....  00000080 00 f4 00 02 00 02 00 03 00 00 00 00 00 00 00  .....3.....  00000090 00 00 00 00 00 00 00 95 00 00 00 02 01 b2 01 00  ...r.....  000000a0 02 00 1d 69 6e 66 69 6d 75 6d 00 04 00 0b 00 00  ...infimum.....  000000b0 73 75 70 72 65 6d 75 6d 13 1e 02 40 00 10 00 3a  supremum...@...  000000c0 80 00 00 03 00 00 00 00 33 86 82 00 00 00 a1 01  .....3.....  000000d0 10 69 de 14 05 80 00 00 02 80 00 00 01 34 31 31  .i.....4111  000000e0 31 20 31 31 31 31 20 31 31 31 31 20 31 31 31  1 1111 1111 1111  000000f0 80 80 13 1e 02 40 00 18 00 3a 80 00 00 04 00 00  ....@.....  00000100 00 00 33 b1 81 00 00 00 af 01 10 69 de 14 4c 80  ...3.....i..L..  00000110 00 00 02 80 00 00 01 34 31 31 31 20 31 31 31  .....4111 1111  00000120 20 31 31 31 31 20 31 31 31 31 80 80 13 1e 02 40   1111 1111....@  00000130 00 20 ff 7c 80 00 00 05 00 00 00 00 33 e8 82 00  . . .....3....  00000140 00 00 bc 01 10 69 de 14 a2 80 00 00 02 80 00 00  .....i.....  00000150 01 34 31 31 31 20 31 31 31 31 20 31 31 31 20  .4111 1111 1111  00000160 31 31 31 31 80 80 00 00 00 00 00 00 00 00 00  1111.....  00000170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ..... </pre> |

| Scenario 5 – Improper Invalidation             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation Steps                           | <pre>\$ sudo mysql USE ospos; DELETE FROM ospos_sales_items WHERE sale_id = 3; DELETE FROM ospos_sales_payments WHERE sale_id = 3; DELETE FROM ospos_sales WHERE sale_id = 3;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Database Storage & Underlying Storage Analysis |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Affected Layer                                 | Server-side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Expected Presence                              | Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Resulting Presence                             | Inconsistent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Steps Taken                                    | <pre>\$ sudo mysql USE ospos; SELECT sale_id, comment FROM ospos_sales ORDER BY sale_id DESC LIMIT 5;  \$ sudo strings /var/lib/mysql/ospos/ospos_sales.ibd   grep 4111 \$ sudo hexdump -C -s 0xfff0 /var/lib/mysql/ospos/ospos_sales.ibd</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Supporting Screenshots                         | <p>Database View (Left) &amp; Underlying Storage (Right), prior to row deletion:</p>  <p>Database View (Left) &amp; Underlying Storage (Right), after row deletion:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Supporting Notes                               | <p>The recovery of deleted PAN data from InnoDB storage files was inconsistent across repeated tests and observations. This indicates that data remanence exists are logical deletion but is influenced by subsequent database activity and behaviour.</p> <p>When deleting numerous rows at once (see Fig. 1), for example, this seems to affect the database's storage behaviour in such a way that securely erases that data from storage. However, when deleting just one row (see Fig. 2), this doesn't seem to cause the database to securely erase it from storage. That said, further activity within the database, while still inconsistent, seems to increase the likelihood of that data being securely erased.</p> <p><i>Fig. 1 – Query, Remove Multiple Rows:</i><br/> DELETE FROM ospos_sales WHERE sale_id &lt; 10;</p> <p><i>Fig. 2 – Query, Remove Single Row:</i><br/> DELETE FROM ospos_sales WHERE sale_id = 3;</p> |

## Appendix C – Illustrative Tables

The PCI DSS Objectives and Requirements table provides an overview of each requirement categorised into the corresponding control objective, as designed by PCI SSC (PCI DSS, 2024).

| Objectives                                          | Requirements                                                                                        |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1 – Build and Maintain a Secure Network and Systems | 1 – Install and Maintain Network Security Controls                                                  |
|                                                     | 2 – Apply Secure Configurations to All System Components                                            |
| 2 – Protect Account Data                            | 3 – Protect Stored Account Data                                                                     |
|                                                     | 4 – Protect Cardholder Data with Strong Cryptography During Transmission Over Open, Public Networks |
| 3 – Maintain a Vulnerability Management Program     | 5 – Protect All Systems and Networks from Malicious Software                                        |
|                                                     | 6 – Develop and Maintain Secure Systems and Software                                                |
| 4 – Implement Strong Access Control Measures        | 7 – Restrict Access to System Components and Cardholder Data by Business Need to Know               |
|                                                     | 8 – Identify Users and Authenticate Access to System Components                                     |
|                                                     | 9 – Restrict Physical Access to Cardholder Data                                                     |
| 5 – Regularly Monitor and Test Networks             | 10 – Log and Monitor All Access to System Components and Cardholder Data                            |
|                                                     | 11 – Test Security of Systems and Networks Regularly                                                |
| 6 – Maintain an Information Security Policy         | 12 – Support Information Security with Organisational Policies and Programs                         |

The CWE-ASVS Control Mapping Table provides an example of how OWASP Top 10 categories, MITRE CWEs and ASVS Control Areas can be mapped to one-another. The example includes those relevant to this study (OWASP, 2025b; MITRE, no date; OWASP, 2025a).

| OWASP Top 10 Category                 | Example CWE                                                  | Relevant ASVS Control Area                |
|---------------------------------------|--------------------------------------------------------------|-------------------------------------------|
| A02 – Security Misconfiguration       | CWE-16 – Misconfiguration                                    | V13 – Configuration                       |
| A04 – Cryptographic Failures          | CWE-319 – Cleartext Transmission of Sensitive Information    | V12 – Secure Communication                |
| A07 – Authentication Failures         | CWE 613 – Insufficient Session Expiration                    | V7 – Session Management                   |
| A09 – Logging and Monitoring Failures | CWE 532 – Insertion of Sensitive Information into a Log File | V16 – Security Logging and Error Handling |



## Appendix D – Expanded Analytical Framework Tables

The expanded misconfiguration matrix table summarises each identified misconfiguration, mapped to its potential manifestation within a POS environment, its potential impact on data persistence, and relevant PCI DSS requirements (PCI DSS, 2024) and MITRE CWEs (MITRE, no date; OWASP, 2025b).

| Misconfiguration                          | Applied in POS Environment                                                              | Potential Impact                                                                                                                | PCI DSS Requirements                                                                                          |
|-------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|                                           |                                                                                         |                                                                                                                                 | MITRE CWEs                                                                                                    |
| Debug mode enabled in production          | POS web application deployed with debugging active                                      | Sensitive transactional data may be written to logs or error outputs, allowing recovery of cardholder data through log analysis | 6 (Secure Development)<br>10 (Logging)<br>3 (Data Protection)                                                 |
|                                           |                                                                                         |                                                                                                                                 | 489 (Active Debug Code)<br>215 (Insertion of Sensitive Information Into Debugging Code)                       |
| Verbose logging                           | Logging configured to capture full request/response data, including transaction details | Long-term storage of sensitive data in log files, increasing the likelihood of forensic recovery through log analysis           | 10 (Logging)<br>3 (Data Protection)                                                                           |
|                                           |                                                                                         |                                                                                                                                 | 532 (Insertion of Sensitive Information into Log File)                                                        |
| Default configurations                    | POS deployed without modifying default settings                                         | Implicit enabling of insecure behaviours such as logging, weak security controls and data retention                             | 2 (Secure Configuration)                                                                                      |
|                                           |                                                                                         |                                                                                                                                 | 16 (Configuration Category)<br>1188 (Initialisation of a Resource with an Insecure Default)                   |
| Lack of encryption at rest                | Stored transaction data (e.g., database and logs) not encrypted                         | Plaintext data can be directly read from storage during forensics analysis                                                      | 3.5 (Protect Stored PAN)                                                                                      |
|                                           |                                                                                         |                                                                                                                                 | 311 (Missing Encryption of Sensitive Data)<br>312 (Cleartext Storage of Sensitive Information)                |
| Improper data invalidation                | Data marked as deleted but not overwritten to ensure complete removal                   | Data remanence allows recovery through data carving                                                                             | 3.2 (Minimum Storage)                                                                                         |
|                                           |                                                                                         |                                                                                                                                 | 226 (Sensitive Information in Resource Not Removed Before Reuse)                                              |
| Backup mismanagement                      | Backups created without encryption or retention limits                                  | Long-term storage of sensitive data in backup files                                                                             | 3 (Data Protection)                                                                                           |
|                                           |                                                                                         |                                                                                                                                 | 530 (Exposure of Backup File to an Unauthorised Control Sphere)<br>311 (Missing Encryption of Sensitive Data) |
| Inconsistent configuration across devices | Different POS terminals configured differently                                          | Unpredictable data storage locations, complicating detection and increasing persistence risk                                    | 2 (Secure Configuration)                                                                                      |
|                                           |                                                                                         |                                                                                                                                 | 16 (Configuration Category)<br>657 (Violation of Secure Design Principles)                                    |
| Session Storage Misuse                    | Browser Local/Session storage contains transactional data                               | Sensitive data may persist client-side beyond intended use                                                                      | 3 (Data Protection)<br>6 (Secure Development)                                                                 |
|                                           |                                                                                         |                                                                                                                                 | 922 (Insecure Storage of Sensitive Information)                                                               |
| Weak access controls                      | Poorly configured database permissions                                                  | Sensitive data accessible to unauthorised users and processes, increasing exposure and recoverability.                          | 7 (Access Control)                                                                                            |
|                                           |                                                                                         |                                                                                                                                 | 284 (Improper Access Control)<br>285 (Improper Authorisation)                                                 |
| Insecure transport                        | POS system transmits transaction details over HTTP rather than HTTPS                    | Payment data may be intercepted or captured in network artefacts through packet analysis                                        | 4 (Encrypt Transmission)                                                                                      |
|                                           |                                                                                         |                                                                                                                                 | 319 (Cleartext Transmission of Sensitive Information)                                                         |

The expanded remanence source table summarises the potential locations of artefacts, mapped to their method(s) of examination, expected data types, and relevant PCI DSS requirements (PCI DSS, 2024), MITRE CWEs (MITRE, no date), and OWASP Categories (OWASP, 2025b).

| Artefact Source       | Location in POS Environment | Examination Method                       | Potential Recoverable Data                  | PCI DSS Requirements                                                                                                                                                |
|-----------------------|-----------------------------|------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                             |                                          |                                             | MITRE CWEs / OWASP (A0)                                                                                                                                             |
| Browser Cache         | Client-side                 | Browser forensic tools, cache inspection | Cached transaction pages, form inputs       | 3.2 (Minimum Storage)<br>6 (Secure Development)                                                                                                                     |
|                       |                             |                                          |                                             | 524 (Use of Cache Containing Sensitive Information)                                                                                                                 |
| Local Session Storage | Client-side                 | Developer tools, storage inspection      | Temporary transaction data                  | 8 (Authentication)                                                                                                                                                  |
|                       |                             |                                          |                                             | A07 (Authentication Failures)<br>922 (Insecure Storage of Sensitive Information)<br>522 (Insufficiently Protected Credentials)                                      |
| Cookies               | Client-side                 | Cookie analysis tools                    | Session IDs, user data                      | 8 (Authentication)                                                                                                                                                  |
|                       |                             |                                          |                                             | A07 (Authentication Failures)<br>315 (Cleartext Storage of Sensitive Information in a Cookie)<br>614 (Sensitive Cookie in HTTPS Session Without 'Secure' Attribute) |
| Application Logs      | Server-side                 | Log file analysis                        | Transaction records, PANs, debug data       | 10 (Logging)<br>3 (Data Protection)                                                                                                                                 |
|                       |                             |                                          |                                             | A09 (Logging Failures)<br>532 (Insertion of Sensitive Information into Log File)                                                                                    |
| Web Server Logs       | Server-side                 | Log file analysis                        | Request/Response data                       | 10 (Logging)<br>3 (Data Protection)                                                                                                                                 |
|                       |                             |                                          |                                             | A09 (Logging Failures)<br>532 (Insertion of Sensitive Information into Log File)                                                                                    |
| Database Storage      | Server-side                 | SQL queries, database dumps              | Stored transaction records, cardholder data | 3 (Data Protection)                                                                                                                                                 |
|                       |                             |                                          |                                             | A02 (Cryptographic Failures)<br>311 (Missing Encryption of Sensitive Data)                                                                                          |
| Temporary Files       | Server-side                 | File system analysis                     | Intermediate transaction data               | 3.2 (Minimum Storage)<br>6 (Secure Development)                                                                                                                     |
|                       |                             |                                          |                                             | 377 (Insecure Temporary File)                                                                                                                                       |
| Backups               | Server-side                 | Backup file analysis                     | Historical transaction data                 | 3 (Data Protection)<br>9 (Physical Access)                                                                                                                          |
|                       |                             |                                          |                                             | 530 (Exposure of Backup File to an Unauthorised Control Sphere)<br>311 (Missing Encryption of Sensitive Data)                                                       |
| Network Packets       | Network Layer               | Packet sniffing tools                    | Possible plaintext payment data             | 4 (Encrypt Transmission)                                                                                                                                            |
|                       |                             |                                          |                                             | A04 (Cryptographic Failures)<br>319 (Cleartext Transmission of Sensitive Information)                                                                               |
| Memory                | Client-side or Server-side  | Memory forensic tools                    | Live transaction data, live session data    | 3.2 (Minimum Storage)                                                                                                                                               |
|                       |                             |                                          |                                             | A06 (Insecure Design)<br>244 (Heap Inspection)                                                                                                                      |

